

Merieme Ait Chabane

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Portfolio: mery-1412.github.io/mery-sPortfolio

SUMMARY

Computer Systems Engineering graduate (Ingénieur d'État and Master's degrees, 2026) with a solid foundation in computer systems, networks, and cybersecurity. Familiar with network and system security, web development, embedded systems, and AI-based projects. Completed a final-year project on real-time detection of network attacks using machine learning. Motivated to deepen my knowledge through graduate study and to contribute to research and innovation in computing.

EDUCATION

Higher School of Computer Science (ESI-SBA), Sidi Bel Abbès, Algeria 2021 – 2026

- Engineer's degree (Ingénieur d'État) in Computer Systems Engineering, Jun 2026
- Master's degree in Computer Systems Engineering, Jun 2026
- Focus areas: computer networks, system and network security, embedded systems, web development, and AI.

EXPERIENCE

Network and Systems Intern, ATM Mobilis (national mobile telecommunications operator) Oran, Algeria - Sep 2024

- Installed and configured operating systems, set up and managed Cisco routers and switches via CLI and web interfaces, and performed network diagnostics and connectivity setup.
- Performed system recovery using Mediat & DLC Boot, and conducted hardware diagnostics (RAM, SSD, CPU) during maintenance procedures.

SELECTED PROJECTS

AI-Based Real Time Detection of PFCP flooding Attacks In 5G Networks *Final Year Project*
Supervised by Dr. Baba-Ahmed Manel and Dr. Ouamri Amine

- Built an end-to-end, real-time intrusion detection framework for PFCP floods on the 5G N4 interface, covering dataset generation, model training, and live deployment with a monitoring dashboard.
- Generated a reproducible dataset on a virtualized Open5GS + UERANSIM testbed.
- Benchmarked supervised (LSTM, BiLSTM) vs. unsupervised (Transformer Autoencoder) models, and selected the Transformer Autoencoder for deployment for its highest recall and lowest latency.
- Deployed the Transformer Autoencoder in a Rust engine for real-time, low-latency detection, with alerts streamed to a live dashboard over WebSocket.
- **Technologies:** Python, TensorFlow/Keras, Rust, ONNX, Scapy, Open5GS, UERANSIM, PFCP/N4, WebSocket, Wireshark, Linux/VMware.

VoteX: Blockchain-Based E-Voting System

4th Year Final Project

- Collaborated on **VoteX**, a hybrid blockchain-based voting dApp that combines the security and transparency of blockchain with the efficiency of traditional web apps.
- Key technologies used: **MERN framework, Next.js, Tailwind CSS, and Solidity.**

Intrusion Detection System

Personal Project

- Designed a **Network Intrusion Detection System (NIDS)** to detect anomalies in network traffic. It captures packets in real time using TShark, detects anomalies with a hybrid system that combines signature-based and anomaly-based detection using the Isolation Forest algorithm, and sends reports to a Flask-based dashboard for visualization.
- Key technologies used: **Python (sklearn, pyshark and related libraries), Flask, TShark, HTML/CSS.**

RELEVANT COURSES

- Computer Networks, Operating Systems, Security of Systems and Networks, Embedded Systems and IoT, Distributed Systems, Software Engineering, Artificial Intelligence.

SKILLS

Networking & Security	Packet Tracer, Wireshark, Nmap, BurpSuite, Autopsy
Operating Systems & Tools	Windows, Linux, Git/GitHub, Docker, VirtualBox, Postman
Programming and Scripting	Python, Java, JavaScript, HTML/CSS
Web Development	MERN (MongoDB, Express.js, React.js, Node.js), Next.js, Tailwind CSS
Embedded Systems & IoT	Arduino, ESP32, ESP8266, Raspberry Pi
Blockchain & Web3	Solidity, Ethers.js, MetaMask, Ganache, IPFS (project experience)
System Design & Modeling	UML, System architecture design
Databases	MySQL, MongoDB (NoSQL)

SOFT SKILLS

Communication, Problem Solving, Teamwork, Organization, Attention to Detail, Adaptability, Work Ethic.

LANGUAGES

English	C1 level (EF SET)
French	B1 level (TCF Online Test)
Arabic	Native

VOLUNTEERING

Scientific and Cultural Clubs

Marketing Department

Contributed to the promotion of club activities and events through content creation, communication, and marketing initiatives. Collaborated with team members to develop engaging content and support the club's visibility.